

SHENYAN ZENG

+86 159 5199 1599 | zengshenyan0308@163.com | samzengsy.cn

EDUCATION

University of California, Santa Barbara

Sep 2021–Mar 2025

Bachelor of Science in Applied Mathematics

- **Relevant Courses:** Numerical Analysis, Linear Algebra, Real Analysis, Optimization, Graph and Network Theory, Probability and Statistics, Computer Science

The University of Hong Kong

Sep 2026–Nov 2027

Master of Science in Engineering in Robotics and Intelligent Systems

- **Planned Coursework:** Robot Modelling, Planning and Control; Robot Sensing and Intelligence; Robotic Systems Integration; Machine Learning and Applications; Human-Robot Collaboration

PUBLICATION

Hao Zhong, Muzhi Zhu, **Shenyan Zeng**, Anzhou Li, Cong Chen, Hua Geng, Duochao Shi, Wentao Ye, Tao Lin, Hao Chen, Chunhua Shen. **Eliciting Complex Spatial Reasoning in MLLMs through Wide-Baseline Matching**. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)

RESEARCH EXPERIENCE

State Key Lab of CAD & CG, Zhejiang University

Hangzhou, China

Research Assistant (Advisor: Prof. Chunhua Shen)

Aug 2025–Present

- **Co-first authored a CVPR 2026 paper** on multimodal spatial reasoning; contributed to the open-source ReasonMatch release with benchmark archives, evaluation scripts, and DCRL training recipe. [\[Project\]](#) [\[Code\]](#)
- Implemented Dynamic Correspondence Reinforcement Learning (DCRL) for Qwen-VL, **boosting absolute F1 by +43.0% (27.5% → 70.5%)** on ReasonMatch-Bench.
- Built scalable automated evaluation pipelines on NAS clusters using vLLM and distributed multi-GPU scheduling for benchmarking models on PointArena and OmniSpatial.
- **Co-first authored an NeurIPS 2026 submission ("SHOTER")**, proposing an innovative multi-stage pipeline that utilizes multimodal large models to convert complex narrative stories into cinematic shot descriptions.
- Designed structural prompts encompassing shot composition and camera planning, successfully guiding closed-source video models to generate temporally coherent cinematic video sequences.

INTERNSHIP EXPERIENCE

Wind Information Co., Ltd.

Nanjing, China

AI Algorithm Engineer Intern

May 2025–Aug 2025

- Fine-tuned multimodal document models (SmoIVLM2, LayoutReader) to achieve **98.8% OCR accuracy** deployed the automated PDF-to-Excel pipeline into a WeChat service “DP 瞬析”.
- Accelerated document AI pipelines by converting RT-DETR/PaddleOCR to ONNX/TensorRT with FP16 quantization, reducing A100 GPU memory usage by 25% and increasing throughput by 3.2×.
- Designed a scalable preprocessing pipeline for **1.25M multimodal samples** (handwriting, formulas, tables) to standardize data for pretraining.
- Developed automated diagnostics (IoU checks, failure visualization), **reducing debugging cycles by 40%** and accelerating model iteration.

Wind Information Co., Ltd.

Jul 2023–Sep 2023

Intelligent Data Analytics Intern

- Annotated **10,000+ structured entries** and standardized DOCX/Excel document merges to improve training dataset consistency.
- Developed Python scripts to automatically parse and aggregate evaluation reports, eliminating manual effort and improving experiment reproducibility.

SKILLS

- **Programming:** Python, C++, PyTorch
- **AI Algorithms:** Transformers, Multimodal Large Models, Diffusion Models, Reinforcement Learning
- **Model Training & Optimization:** LoRA / PEFT, Distributed Training, Prompt Engineering
- **Systems & Deployment:** vLLM, TensorRT, ONNX, CUDA, Multi-GPU Inference
- **Development & Tools:** Linux, Git, Docker, AI-assisted development tools
- **Languages:** English (Fluent), Mandarin (Native)